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Technical Note



Methylation-based forensic age estimation in blood, buccal cells, saliva and semen: A comparison of two technologies

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ABSTRACT

Forensic age estimation of stain donors through DNA methylation has been intensively studied in recent years. To date, there are many published age estimation tools which are based on technologies including pyrosequencing, minisequencing, or MPS. With the implementation of such tools into routine forensic casework in many laboratories worldwide, there is a need for thorough evaluation and performance comparison. In this study, we tested published age estimation tools that are based on either minisequencing or MPS on four body fluids (blood, saliva, buccal cells and semen). All samples were analyzed with both technologies and the age estimates were compared. Biological replicates were taken from ten (blood, saliva, buccal cells) or 12 individuals (semen) to assess the reproducibility of each tool. Our study demonstrates high accuracy in estimating chronological age for various body fluids using both technologies, except for semen. The mean absolute errors (MAEs) ranged from three to five years for blood, saliva and buccal cells, while semen exhibited a higher MAE of seven to eight years. Despite the overall good performance for blood, saliva, and buccal cells, significant discrepancies were observed for some individuals both between the two technologies or when compared to their chronological age. Conclusively, we demonstrated that forensic age estimation tools based on two different technologies are similarly accurate for blood, saliva and buccal cells, while the semen tools need some adjustments before implementation into forensic casework. Our results could be helpful in the decision-making process for laboratories seeking to newly establish an age estimation workflow.

1. Introduction

Methylation-based forensic age estimation of a donor from crime scene stains has been a widely studied topic within forensic DNA phenotyping in recent years [1,2]. Its main purpose is to provide investigative leads on a potential perpetrator when traditional approaches like STR profiling do not produce a hit in a DNA database [1,3]. By providing an estimate of the chronological age of the stain donor, the police can narrow down the pool of suspects. Although there have been attempts to use other markers like RNA, telomere length or aspartic acid racemization for the same purpose [4–7], DNA methylation is currently the method of choice for this application in forensic laboratories [8].

Based on Horvath's original multi-tissue epigenetic clock [9], forensic age estimation tools have exploited age-dependent methylation changes at specific CpG sites to predict the age from biological material of a donor [10,1,11–13]. Contrary to Horvath's and other early developed epigenetic age clocks [9,14,15], forensic tools often investigate fewer markers [10,13,16,17] to make the tool suitable for forensic use where DNA quality and quantity are a commonly limiting factor [18]. In addition, most forensic age estimation tools are specifically designed for a particular body fluid in order to achieve the highest possible estimation accuracy [3]. Such forensically relevant body fluids include blood, saliva, semen, bones, fingernails, and others [10,12,19,20].

Many different tools, comprising a workflow to analyze methylation patterns at selected target CpG sites and a statistical model for age calculation, have been developed for methylation-based age estimation in recent years. They are based on different technologies including pyrosequencing, minisequencing (e.g. SNaPshot™), microarrays and massively parallel sequencing (MPS) [3,10,11,16,21]. Although still in its very early stages in the field of forensic genetics, nanopore sequencing has also recently been proposed as another potential technology for methylation-based applications including age estimation [22]. Most of the tools developed so far for each of these technologies perform similarly well in terms of age estimation accuracy, usually with

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a mean absolute error (MAE) in age estimation between three and five years within the respective study cohorts. Independent of the technology, methylation-based age estimation is more difficult in elderly individuals, especially above 60 years of age, where the MAEs usually increase. This is likely explained by the increasing influence of genetic and environmental factors that cause greater fluctuations in methylation patterns [10,23,24].

When comparing the underlying technologies themselves, several differences can be observed. For instance, pyrosequencing, microarrays and MPS are quantitative methods, whereas SNaPshot is semi-quantitative. This can lead to differences in measured methylation values when compared between these technologies [16]. Consequently, statistical models that are built on data from one technology cannot be applied to data generated with another technology, even if the same target CpG sites are being investigated [1]. An additional difference between the technologies mentioned is the practical handling time in the laboratory. For example, SNaPshot is a very straight-forward and quick 1-day protocol while MPS needs considerably more time (at least two working days) from DNA extraction to analyzed results.

Among the existing forensic age estimation tools, one that is based on SNaPshot technology was developed in 2019 by Jung et al. [11]. It examines five different target CpG sites in five different genes (*ELOVL2*, *KLF14*, *MIR29B2C*, *TRIM59* and *FHL2*) for age estimation from blood, buccal cells or saliva. The underlying study cohort consisted of 150 blood samples, 150 saliva samples and 148 buccal cell samples from 304 Korean participants between the age of 18 and 74 years. The same target sites are used for all three body fluids, but the models created by linear regression are different depending on the investigated body fluid. For semen samples, the same group also developed a SNaPshot tool that targets three CpG sites, namely cg12837463 and one CpG site each in the genes *TTC7B* and *NOX4* [25]. The study cohort consisted of 94 male Korean participants between the age of 20–73 years.

Another forensic age estimation tool which is based on MPS was developed in 2020 by the VISAGE Consortium [10]. It is a combined tool for age estimation from either blood, buccal cells or bones. For both blood and buccal cell samples, 160 presumably European participants aged between one and 80 years contributed samples, respectively. Eight gene amplicons are sequenced simultaneously (*ELOVL2*, *KLF14*, *MIR29B2C*, *PDE4C*, *EDARADD*, *FHL2*, *TRIM59* and *ASPA*) for the analysis of different target CpG sites depending on the investigated body fluid. A statistical model for each investigated body fluid was generated by linear regression. The VISAGE group also developed an MPS-based tool for semen samples targeting six CpG sites in five different genes (*SH2B2*, *EXOC3*, *IFITM2*, *GALR2* and *FOLH1B*) [12]. This study was based on a cohort of 288 semen samples from European participants between 24 and 58 years of age.

In our study, we compared the performance of the above-mentioned age estimation tools for blood, saliva, buccal cells and semen. We tested each body fluid with the corresponding SNaPshot- and MPS-based tool and evaluated the age estimations both in comparison to the chronological age of the stain donor and between tools. We aimed to show whether different age estimation tools, that are based on different technologies, target genes, statistical models, cohort sizes and populations, perform equally well in terms of estimation accuracy and reproducibility on the same samples in our laboratory. In doing so, we sought to provide guidance to other forensic laboratories that wish to implement an age estimation workflow.

2. Methods

2.1. Study cohort

For this study, we collected blood, saliva and a buccal swab samples from 37 individuals on the same day (n male = 10, n female = 27, age range = 18.29–61.82 years, see Supplementary table 1). From 10 out of these 37 individuals, biological replicates for all three body fluids were collected on another day.

In addition, we collected semen samples from 47 males (age range = 21.07–59.10 years). These individuals were different participants than those donating blood, buccal swabs and saliva. 12 out of the 47 semen donors provided an additional sample on a different day. Two individuals in the semen cohort were self-reported azoospermic donors, but their samples were treated the same way as all other samples.

Chronological age, sex and biogeographical background (self-reported based on each individual's knowledge of their ancestors) were enquired from all participants. This study was approved by the ethics committee from the University of Zurich (CEBES_2021–11a) and all participants provided an informed consent.

2.2. Sample collection

Peripheral blood was taken from a finger with a finger-prick lancet and was directly collected with a cotton swab (milian®, Bremgarten, Switzerland) until the swab was soaked. Each participant also swabbed their inner cheek with a cotton swab for approximately 30 sec to obtain the buccal cell sample. All swabs were air-dried for at least one hour before storage at room temperature until DNA extraction. Spit saliva was collected in an Eppendorf tube and stored at -20 °C. For DNA extraction, 100 µl of thawed saliva were pipetted directly into the lysis buffer.

For the semen samples, participants collected semen themselves in 50 ml falcon tubes, which were stored at -20 °C until further processing. For DNA extraction, 50 µl of thawed semen were pipetted directly into the lysis buffer.

2.3. DNA extraction, DNA quantification and bisulfite conversion

All samples were extracted with the QIAamp® DNA Investigator kit (Qiagen, Hilden, Germany) either manually according to the "Body fluid stains" protocol or with a QIAcube instrument (Qiagen) using the "Surface and buccal swab" protocol. DNA quantity was measured with the Quantus Fluorometer (Promega, Madison, USA). Bisulfite conversion was subsequently performed on 20 µl of extracted DNA with the MethylEdge® Bisulfite Conversion System (Promega) and eluted in 10 µl elution buffer. For calculation of the estimated DNA concentration after the bisulfite conversion, the following formula was used: (DNA concentration [ng/µl] * 20 µl)/10 µl. The same bisulfite converted DNA of each sample was used for both downstream workflows.

2.4. SNaPshot workflow

The SNaPshot reaction was performed according to the protocol published in Jung et al. [11] (for blood, saliva and buccal cells) and Lee et al. [25] (for semen). The optimal input amount of bisulfite converted DNA into the initial PCR is between 10 and 20 ng. If the input was lower than 10 ng, the maximum PCR input amount of 4 µl was used for all body fluids. Mean input amounts per body fluid into the intial PCR are listed in Supplementary table 2. In each run, a 0 % methylated control and a 100 % methylated control from the Human Methylated and Non-Methylated DNA set (Zymo Research, Irvine, USA) was used to assess the bisulfite conversion efficiency. If the methylation percentages at the target sites for the 0 % standard were between 0 % and 5 % and the methylation percentages of the 100 % standard were between 95 % and 100 %, the conversion was deemed successful. SNaPshot products were run on a SeqStudio Genetic Analyzer (ThermoFisher Scientific, Waltham, USA) instead of an Applied Biosystems® 3130 Genetic Analyzer (ThermoFisher Scientific) as in the original publications. For the final age estimation, we used the model for the blood, saliva and buccal cell samples as published in [11], while for the semen samples we used the model as published in [26].

2.5. MPS workflow

The MPS workflow was performed as published in Woźniak et al. [10] (for blood, saliva and buccal cells) and Pisarek et al. [12] (for semen). 40 ng or more is the optimal bisulfite converted DNA input for all body fluids in the multiplex PCR reaction. If an input of 40 ng could not be reached, the maximum volume of 8 µl of bisulfite converted DNA was used for all body fluids. Mean input amounts into the intial PCR are listed in Supplementary table 2. Paired-end sequencing of 201 cycles was performed for all samples with a MiSeq FGx Reagent Micro kit (Qiagen) on a MiSeq FGx instrument (Illumina, San Diego, USA). Sequence alignment from the fastq files was done with a custom Python script using the bwa aligner for bisulfite-treated sequences [27]. Bam files were created and indexed with Samtools [28]. Read counts were extracted based on unmerged reads from the bam files. For the calculation of the age estimates, a dongle containing the VISAGE software, that is available on request from the VISAGE consortium, was used. Run quality was assessed with custom R scripts checking for: 1) bisulfite conversion efficiency $\geq$ 90 %, 2) base misincorporation rate $\leq$ 2 %, 3) read depth of at least 1000 reads at each target site, and 4) a visual inspection of the alignment in the Integrative Genomics Viewer (IGV) [29].

2.6. Data analysis

Data analysis was performed in R. Beta values at the target sites were calculated by dividing the peak heights (SNaPshot) or reads (MPS) corresponding to the methylated cytosines by the sum of the methylated and unmethylated peak heights (SNaPshot) or reads (MPS). Age estimates were generated with the respective statistical models [1,10–12,25,26]. One person was excluded from all analyses except the investigation of biological replicates due to a SNP in one of the target CpG sites. PCA plots on the beta values of the target sites were generated with the R function *prcomp* [30] (see Supplementary figure 1). As the MPS tool does not have an explicit statistical model for saliva samples, the model and target sites for buccal cells were used on the saliva samples.

Statistical comparisons and graphical representations were performed with the R packages *stats*, *ggplot2* and *tidyverse* [30–32].

3. Results

3.1. Age estimation comparison

Age was estimated from all samples using both SNaPshot- and MPS-based workflows. Correlations between the estimated and the chronological age were high ($R^2 > 0.8$) for saliva and buccal cells using both tools. While blood samples also showed correlations of $R^2 > 0.8$, they had a slightly higher overall variance. The estimations of all semen samples showed a considerably weaker correlation with chronological age ($R^2 = 0.57$ for SNaPshot and $R^2 = 0.20$ for MPS), especially in individuals older than 40 years of age. When excluding samples with a chronological age above 40 (Fig. 1), the correlation only slightly increased. Overall, the age estimates for semen in persons under 40 years of age (u40) were underestimated with SNaPshot and overestimated with MPS. A per sample comparison of the SNaPshot and MPS estimates was performed as well (Supplementary table 3).

One individual was excluded from the above-mentioned analysis because she has a heterozygous SNP in the marker *KLF14* at position chr7:130734355. The SNP was identified visually from the MPS sequencing results in IGV (Supplementary figure 2). This is a target CpG site in the SNaPshot tool, and in the saliva and buccal cell models in the MPS tool. The SNP might have affected the age estimation of the SNaPshot tool in all three body fluids, although this was not consistently observed across all body fluids and replicates. It could also have affected the estimations of saliva and buccal cells in the MPS tool, although apparently not as much as the SNaPshot estimates (Fig. 2).

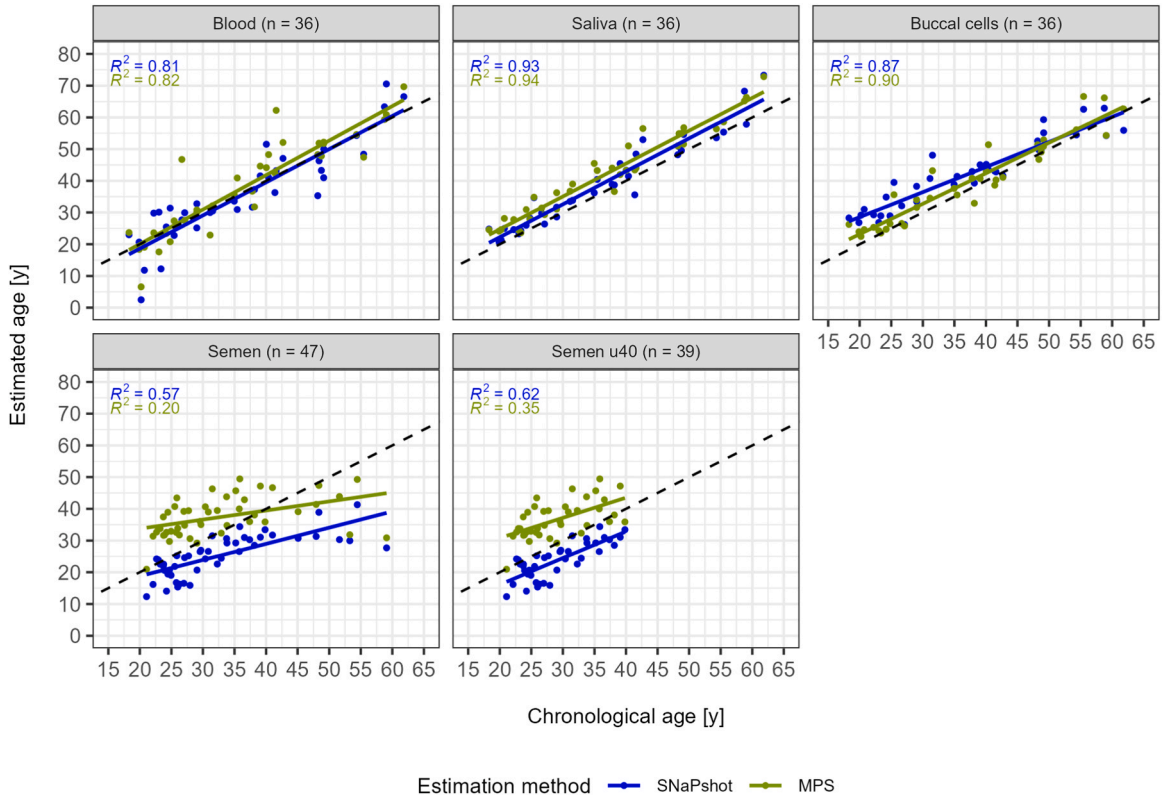


Fig. 1. Comparison of the age estimates between the SNaPshot and MPS tool per body fluid. Mean absolute error in years (MAE) SNaPshot: 4.94 (Blood), 3.37 (Saliva) and 5.75 (Buccal cells), 7.52 (Semen), 5.50 (Semen u40); MAE MPS: 4.43 (Blood), 5.40 (Saliva), 3.55 (Buccal cells), 8.24 (Semen), 7.84 (Semen u40).

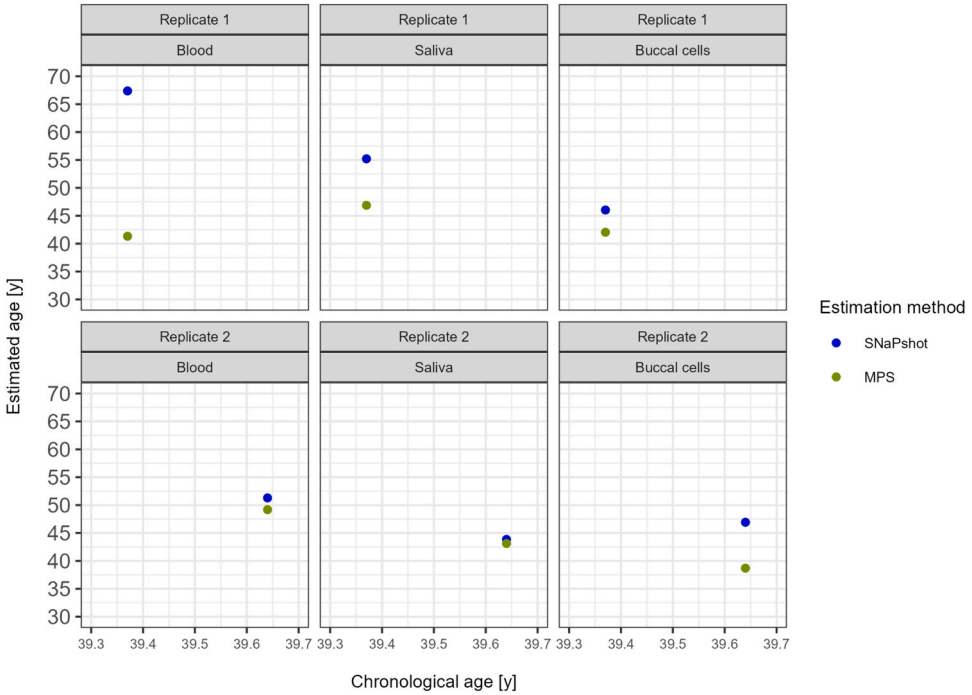


Fig. 2. Age estimation results from the individual with a SNP in *KLF14* at position chr7:130734355.

The most common parameter to assess age estimation accuracy across a study cohort is the mean absolute error (MAE), that represents the mean of all absolute differences between the estimated and chronological age within a study cohort. While this allows for a quick assessment of the accuracy of an age estimation tool, other parameters such as the root mean square error (RMSE) or an in-detail analysis of single observations is required to get a complete picture of a tool's performance. The MAE and RMSE values for blood, saliva and buccal cells were comparable between the two tools and across all three body fluids (Table 1). However, it is noteworthy that the absolute estimation errors across all buccal cell samples were statistically significantly greater than the ones for the saliva samples with the SNaPshot tool (p-value = 0.003, Wilcoxon test; Fig. 3). With the MPS tool, the absolute estimation errors for the saliva samples were slightly greater than those for the blood samples (p-value = 0.02, Wilcoxon text) and statistically significantly greater than the absolute errors for the buccal cell samples (p-value = 0.007, Wilcoxon test).

Comparing the absolute errors for each body fluid between the two tools, the blood sample errors were not statistically significantly different (p-value = 0.263, Wilcoxon test). However, the errors of both the saliva and buccal cell samples were statistically significantly different between the two tools (p-value (saliva sample error): 0.002; p-value (buccal cell sample error): 0.007, Wilcoxon test).

Both tools showed a comparably larger MAE and RMSE value for the entire semen cohort (Table 1). When excluding all samples with a chronological age above 40, the MAE of the SNaPshot tool was reduced by approximately two years, while the MAE of the MPS tool declined by less than one year (Table 1). Two semen donors were self-reported azoospermic and their methylation profiles were investigated in a PCA plot (Supplementary figure 1). As their age estimates did not deviate more than the other estimates in the semen cohort, they were included in all further analyses. As buccal cell and saliva samples are a very similar sample type and were already estimated with the same model and target sites in the MPS-based tool, the statistical model for saliva samples from the SNaPshot tool was also applied to the SNaPshot buccal cell data. A direct comparison of the buccal cell and saliva model indicated improved age estimation when the saliva model was applied to the buccal cell samples (Fig. 4).

Table 1
MAE/RMSE values for all body fluids [y].

	SNaPshot		MPS	
Body fluid	MAE	RMSE	MAE	RMSE
Blood (n = 36)	4.94	6.41	4.43	6.67
Saliva (n = 36)	3.37	4.58	5.40	6.24
Buccal cells (n = 36)	5.75	6.87	3.55	4.78
Semen (n = 47)	7.52	9.72	8.24	9.93
Semen u40 (n = 39)	5.50	6.46	7.84	9.03

3.2. Biological replicates

Biological replicates were collected from ten participants for blood, saliva and buccal cell samples and from 12 participants for semen samples. Both replicates were compared with each other and against the chronological age of the corresponding person (Table 2). At least half of the saliva and buccal cell samples had replicate differences of less than three years, even if they did not correspond well with the chronological age (Fig. 5). For the blood samples, estimated with both tools, and the MPS estimates of the semen samples, bigger differences between replicates were observed, consistent with the previously observed bigger variance in the entire blood and semen cohorts.

3.3. Individual observations

Even though the ages of the study cohort as a whole were estimated reasonably accurately for blood, saliva and buccal cells, some observations for single cases differed considerably from the chronological age. The following examples in Fig. 6 demonstrate three individuals for whom 1) the estimated and chronological age correspond very well across all body fluids and both tools (ID23), 2) either the SNaPshot tool or the MPS tool showed a greater estimation error in one body fluid (ID24), and 3) the estimation in one body fluid was considerably off for both tools (ID25). It is noteworthy that ID23 and ID24, the reasonably accurate examples, were much older (54 and 48 years, respectively) compared to ID25, the inaccurately estimated example (20 years).

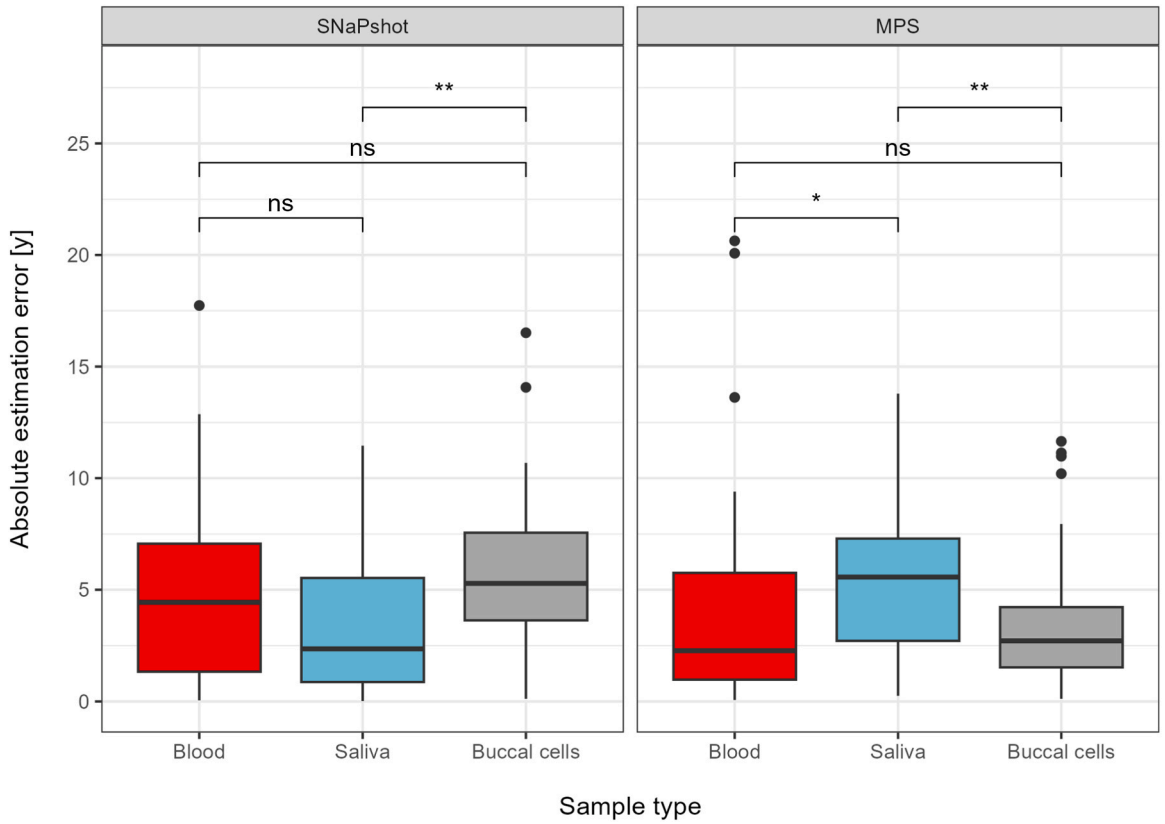


Fig. 3. Comparison of the absolute estimation errors between body fluids [y]. All three body fluids were collected per person on the same day. Semen samples are not shown because they were collected from different individuals.

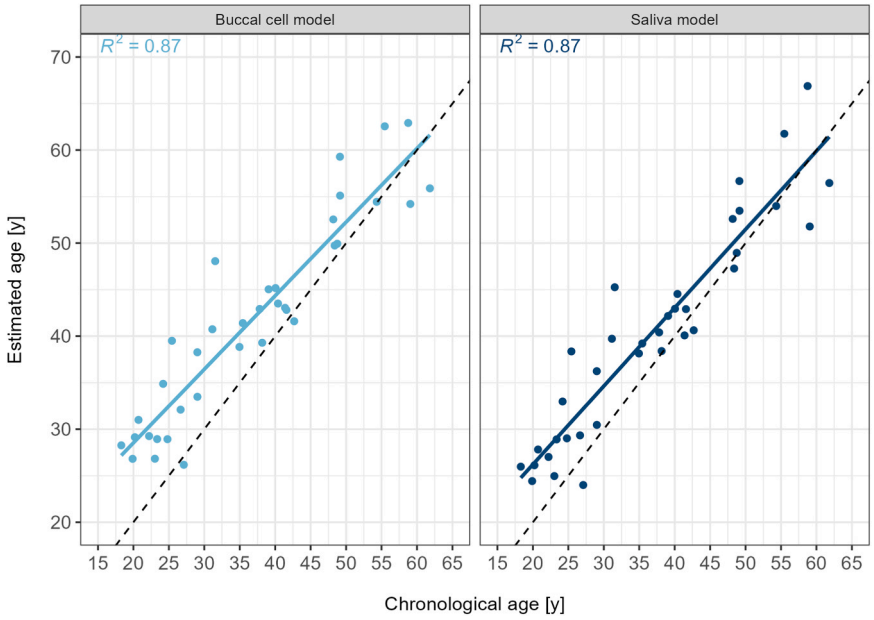


Fig. 4. Model comparison on buccal cell samples with the statistical models for buccal cells and saliva for the SNaPshot tool. MAE in years: 5.73 (Buccal cell model), 4.72 (Saliva model).

4. Discussion

This study aimed to compare the accuracy of published tools for quantitative methylation-based age estimation from donors of crime scene stains using two different technologies. It was intended to demonstrate whether tools based on different technologies, target genes, statistical models, cohort sizes and populations are equally practicable for forensic casework and can be used interchangeably or as an independent confirmation in a laboratory where both workflows are established. Furthermore, it strived to highlight some caveats for the use of either of the investigated tools and for the application of methylation-based age estimation in a forensic setting in general. Although the

Table 2
Mean absolute age estimation difference between biological replicates per body fluid and technology [y].

Body fluid	SNaPshot	MPS
Blood (n = 10)	7.21	4.55
Saliva (n = 10)	3.99	2.74
Buccal cells (n = 10)	2.16	2.54
Semen (n = 12)	2.70	5.25

results of our study demonstrate some limitations in both tools, it is remarkable how well they perform considering the different laboratory and different kits used in comparison to the original studies. However, it still suggests that these tools may not be directly applicable in other laboratories, and that statistical modelling might have to be adapted to the workflow of the respective laboratory.

Our results clearly showed that all investigated age estimation tools performed comparably well in terms of age estimation accuracy on the same samples, based on the MAEs and mean differences of age estimates from the SNaPshot and MPS tools per sample. Only for the semen cohort, a much bigger mean difference in estimates between the two tools was observed although the MAEs were comparable. This is likely due to the generally less accurate performance of the semen tools in comparison to the other body fluids and will be discussed further below. In this study, the same bisulfite converted DNA was used for both downstream workflows ensuring that any difference in age estimation accuracy was not caused by upstream variations in DNA extraction or bisulfite conversion.

For blood samples, there was no statistically significant difference between the estimation errors of both tools, and the MAEs (MAE SNaPshot tool = 4.94 years, MAE MPS tool = 4.43 years) were also very similar (within 1.5 years of difference) to those of the published study cohorts (published MAE SNaPshot tool = 3.48 years, published MAE MPS tool = 3.2 years) [10,11]. However, the estimates from blood samples had a slightly higher variance than those from saliva and buccal cell samples (Fig. 1). This could be explained by the generally lower amount of DNA retrievable from forensically relevant blood stains due to the reduced number of nucleated cells (Supplementary table 2). As commonly known in methylation-based age estimation applications, the less DNA is used for the analysis, the greater the risk of introducing stochastic variation in the observed methylation patterns [1,33,34]. For the blood sample cohort, this could have led to sporadically greater estimation errors and hence partially explain the larger scatter of the blood estimates in general.

For the saliva samples, the MAE of 3.37 years obtained with the SNaPshot tool is very similar to the published MAE (3.55 years) [11]. When comparing the SNaPshot MAE with the MPS estimates, the latter are statistically significantly greater with an MAE of 5.4 years. As the MPS tool does not have a published statistical model specifically designed for saliva samples, the target sites and the model for buccal cells was used to calculate age from the saliva samples. Therefore, it is not surprising that these estimates differ more from the chronological age of the sample donor, as the statistical model was not trained on saliva samples. Although saliva and buccal cell samples are very similar sample types, it has been previously reported that the number of leukocytes in saliva can vary considerably between and within individuals [35]. In comparison, buccal cell samples are expected to contain mostly buccal epithelial cells, whose methylation patterns differ from those of leukocytes [36]. Therefore, depending on the ratio of buccal cells to leukocytes in saliva, the measured methylation values are more or less similar between the two sample types, giving rise to smaller or larger age estimation errors.

The MAE for buccal cell samples differed more from the published MAEs for the SNaPshot tool than in the blood and saliva samples (MAE SNaPshot tool = 5.75 years), but was almost equal (MAE MPS tool = 3.55 years) to the published MAE for the MPS tool (published MAE SNaPshot tool = 4.29 years, published MAE MPS tool = 3.7 years).

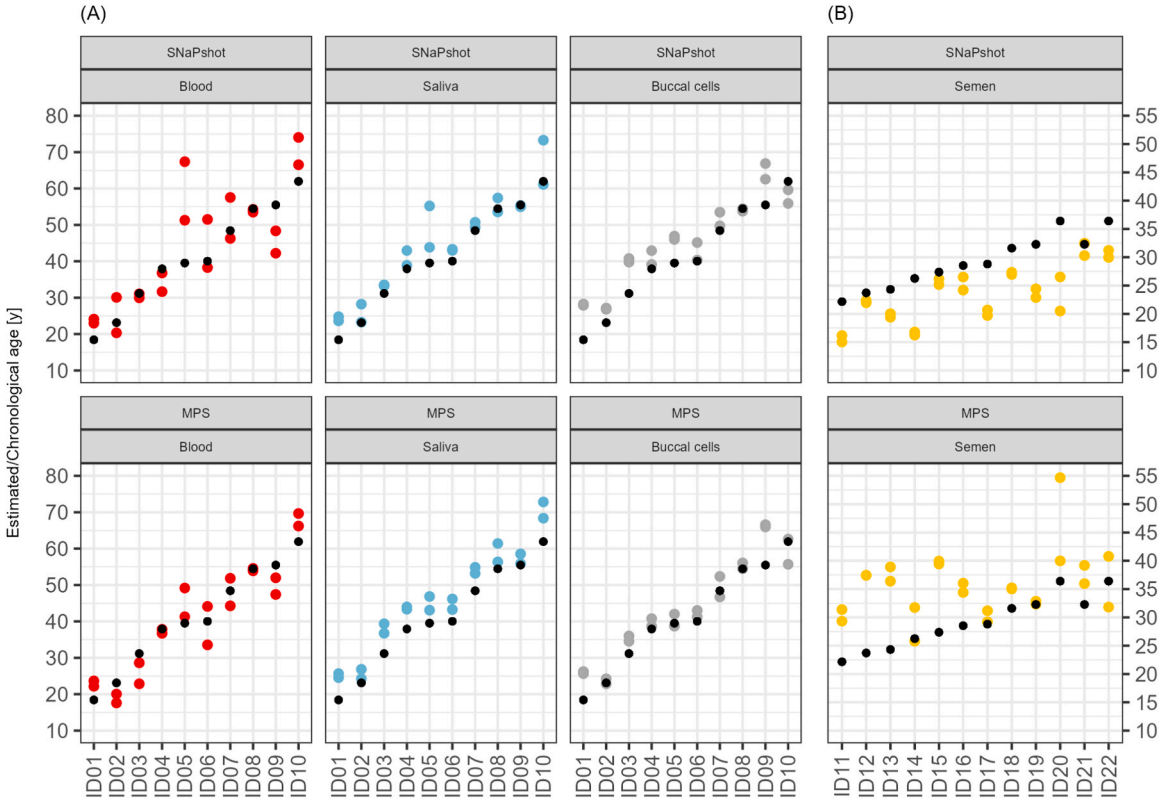


Fig. 5. Biological replicate measurements for (A) ten individuals (ID01 - ID10), that donated blood (red dots), saliva (blue dots) and buccal cells (grey dots), and (B) 12 individuals (ID11 - 22), that donated semen (yellow dots), for the SNaPshot and MPS tools. Black dots indicate the chronological age of the respective individual.

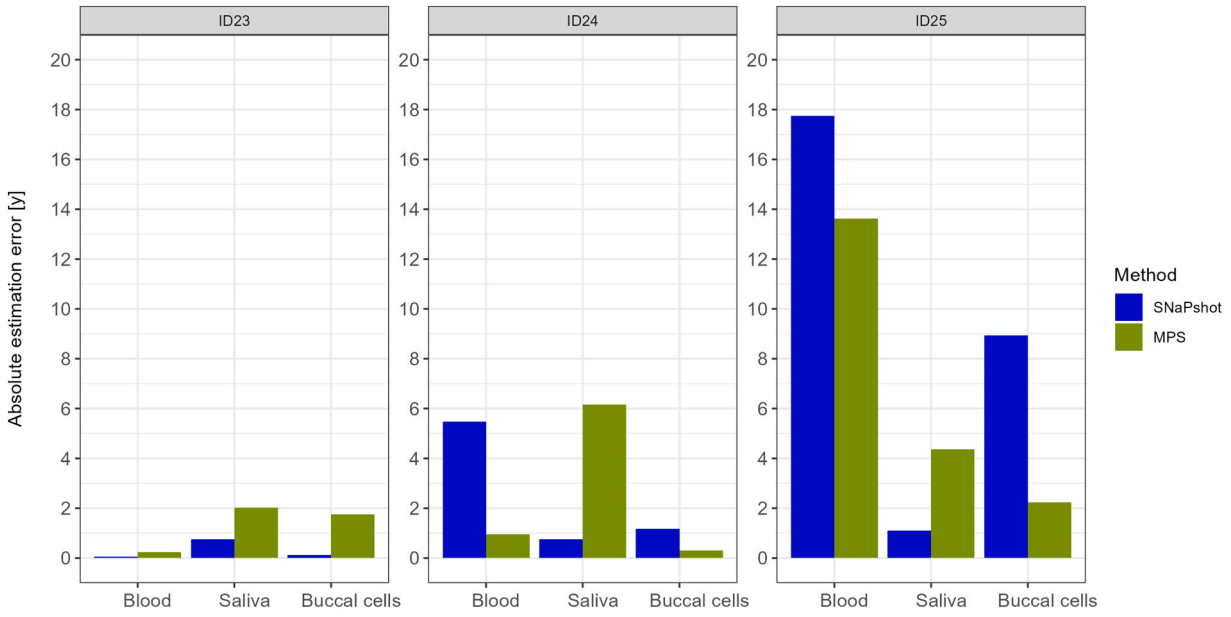


Fig. 6. Absolute age estimation errors of three exemplary persons [y]. For ID23, accurate age estimates were observed with both tools across all body fluids. For ID24, the SNaPshot estimate for blood was less accurate than the MPS estimate, while for saliva, the MPS estimate showed a greater error than the SNaPshot one. Buccal cells of ID24 were estimated accurately with both tools. All estimates for ID25 were considerably off compared to the chronological age except for the SNaPshot estimate of saliva and the buccal cell estimate from the MPS tool.

Interestingly, the SNaPshot estimates improved considerably when the statistical model for saliva was used (MAE = 4.72 years) instead of the one for buccal cells. This again demonstrates the close relationship between these two sample types, but also highlights the varying cellular composition in saliva and suggests that further evaluation of these models may be required.

Apart from the comparability of age estimates of the two tools for blood, saliva and buccal cell samples, our results also suggest that estimates from different body fluids of the same person are generally very similar (Fig. 3). This is mostly due to a well selected set of target sites for each body fluid, and a well-built statistical model. Interestingly, with the SNaPshot tool the median estimation values of the three body fluids differed, possibly indicating some general differences between the body fluids. In the MPS estimates, the median of the blood and buccal cell samples were comparable, and the higher median of the saliva samples can likely be explained by the lack of a dedicated statistical model for saliva samples (Fig. 3). Overall, this still suggests that body fluids collected on the same day can be used interchangeably for age estimation and should yield roughly the same results. In line with this, we found that inaccuracies in age estimates for single individuals are mostly reflected across all body fluids (Fig. 6). Biological reasons such as different life style, environmental exposure or the biogeographical background [3,37], affecting methylation in all body fluids, could be responsible for such observations, although this was not assessed in this study. Only in one individual of Scandinavian origin, we found a clear biological explanation for a greater estimation error. This person has a SNP in one of the target CpG sites of both the SNaPshot and MPS tools. Interestingly, this SNP has been previously reported in another publication on forensic age estimation [20] and has an allele frequency in Northern Europeans that is almost twice as high as in all other Europeans (0.0070 and 0.0038, respectively, according to gnomAD). Therefore, this target site might not be the most suitable one and may need to be replaced, especially for use in regions with a large Scandinavian population. This is particularly important when no sequencing-based tool is used, as this SNP would have remained undetected with a technology like minisequencing.

The semen cohort, which consisted of different individuals than the blood, saliva and buccal cell cohorts, was estimated worse than the other three body fluids. An MAE of 7.52 for SNaPshot and 8.24 for MPS indicates that these models, and potentially the chosen target sites, need to be adjusted before they can be implemented into forensic casework. Interestingly, the MAEs in our study cohort are also considerably larger (more than two years difference) than in the published data sets (MAE SNaPshot = 5.4 years, MAE MPS = 5.10 years). As it is known that defects of methylation pattern maintenance become more common with increasing age [38], it was investigated whether a subset of our cohort consisting of all individuals younger than 40 years of age yielded better estimations than the entire cohort. Excluding all individuals above 40 (n = 8) resulted in an MAE of 5.50 years in the SNaPshot estimates which can be considered as an acceptable overall error. However, the MAE for the MPS estimates did not differ considerably between the subset and the entire cohort, namely 7.84 years vs. 8.24 years, both of which are high errors in a forensic setting. This again emphasises the need for model re-training, analysis of a larger sample size and further investigation of methylation patterns in semen samples for forensic applications. Although even the original epigenetic clock by Horvath [9] did not perform well on semen samples, other target sites, like some used in other age estimation tools (e.g. [39,40]), might be more suitable for this sample type. Finally, it is also important to consider biological reasons that may contribute to difficulties in accurately estimating age from semen like azoospermic donors [41]. Our study cohort included two azoospermic donors whose methylation pattern clustered slightly outside of the remaining semen samples in a PCA plot for both the MPS and SNaPshot target sites (Supplementary figure 1). However, this did not affect the age estimation accuracy in a significant way compared to the other semen samples. Still, it might be worth investigating this effect in more detail in future studies.

Another aspect investigated in this study was the reproducibility of age estimates for samples collected from the same individuals on different days. Because of the susceptibility of methylation to environmental factors, it is expected that estimates will show some variation when sampling at different time points. However, this study found that biological replicates were reasonably reproducible with all investigated tools and across the four body fluids. This is not very surprising, as the sampling time points were not more than three months apart and methylation patterns are not expected to vary greatly within this short

time interval. The mean estimation differences between the replicates ranged from 2 to 5 years for all body fluids and tools, except for the blood replicates estimated with SNaPshot. This is in line with the previously observed greater variation in blood estimates and could again be explained by the relatively small amount of DNA forensically relevant blood samples can provide. While the 2- to 5-year variation between replicates could be improved, this is still considered an overall high reproducibility within samples that were least possibly controlled to mimick casework samples as closely as possible (no assessment of potential disease, stress, other environmental factors at sample collection). This is highly relevant for forensic applications as only reasonably reproducible methods should be used for casework. Another relevant aspect for forensic applications, which was not investigated in this study, is the analysis of samples with a long storage time. As it has been shown in the literature that DNA methylation is generally very stable under various conditions [42], it is assumed that methylation-based age estimation is also reproducible over time. Still, it might be relevant for forensic casework to investigate this in more detail and in a forensic setting in future studies.

With regard to casework applications of methylation-based age estimation, this study also highlights a crucial aspect that is rarely discussed in publications. While the overall performance of all investigated age estimation tools is very good in terms of estimation accuracy for all body fluids with the exception of semen, this is not always true for individual observations. In this study three individuals were used to illustrate that some individuals can be estimated consistently accurately with both tools while others can deviate up to 18 years from the chronological age, either for one or both tools. This is an important consideration for casework applications. Even if a tool has been shown to perform accurately across a large sample cohort, this does not necessarily ensure accurate estimates for individual cases. Again, this highlights that the MAE, which is the mean error within a given study cohort, does not represent the maximum error to be expected when using the respective age estimation tool.

In conclusion, this study showed that two selected SNaPshot- and MPS-based tools for methylation-based age estimation in blood, buccal cells and saliva, namely those published by Jung et al. [11] and Wozniak et al. [10], perform almost equally well across three different body fluids. While the investigated semen tools [12,25] performed worse in comparison to the chronological age of each sample donor, the estimates between the two tools were at least reasonably comparable. Overall, this is highly relevant information for forensic laboratories, as the two technologies differ in terms of required equipment, costs and sample handling time. If a laboratory does not have a sequencing machine, a SNaPshot assay can be an equally good alternative for age estimation analysis. Additionally, the SNaPshot workflow is cheaper and requires less time from bisulfite treatment to final results (approximately one day) than MPS (approximately two days), which can be a critical aspect for casework. On the other hand, a sequencing-based approach offers the advantage of being a quantitative method in comparison to the semi-quantitative SNaPshot approach. This could potentially lead to more robust results in a larger sample cohort, although so far, our study does not support such a suggestion. Still, the MPS-based approach allows for more quality control measures and therefore easier trouble-shooting if necessary. When both technologies are available, our study further shows that it might be possible and useful for forensic applications to apply both SNaPshot- and MPS-based tools for an independent result confirmation.

Finally, our study showed that the tested tools for blood, saliva and buccal cells perform very well on both technologies, while the two tools used for semen age estimation are not yet accurate enough for an implementation into routine forensic applications. We suggest that age estimation works equally well on different body fluids with the two different technologies and different investigated tools. These findings may help other forensic laboratories in making decisions before implementing an age estimation workflow.

CRediT authorship contribution statement

Charlotte Sutter: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yael Marti:** Writing – review & editing, Formal analysis, Data curation. **Cordula Haas:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Jacqueline Neubauer:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.forsciint.2024.112325.

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